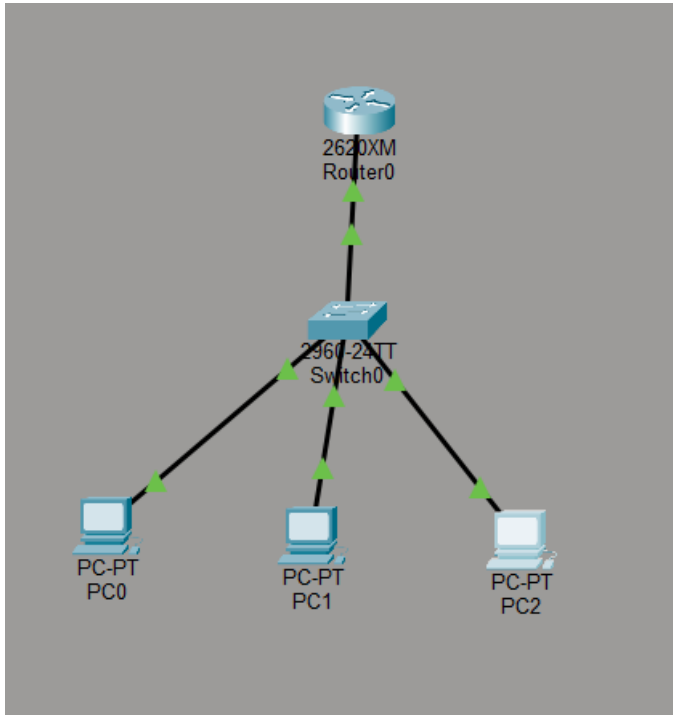


Q-Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Configure a DHCP server to automatically assign IP addresses to at least three devices and verify that each device receives a unique IP address from the configured range.

Network Topology



Router DHCP Configuration:

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.20
Router(config)#ip dhcp pool LAB_POOL
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

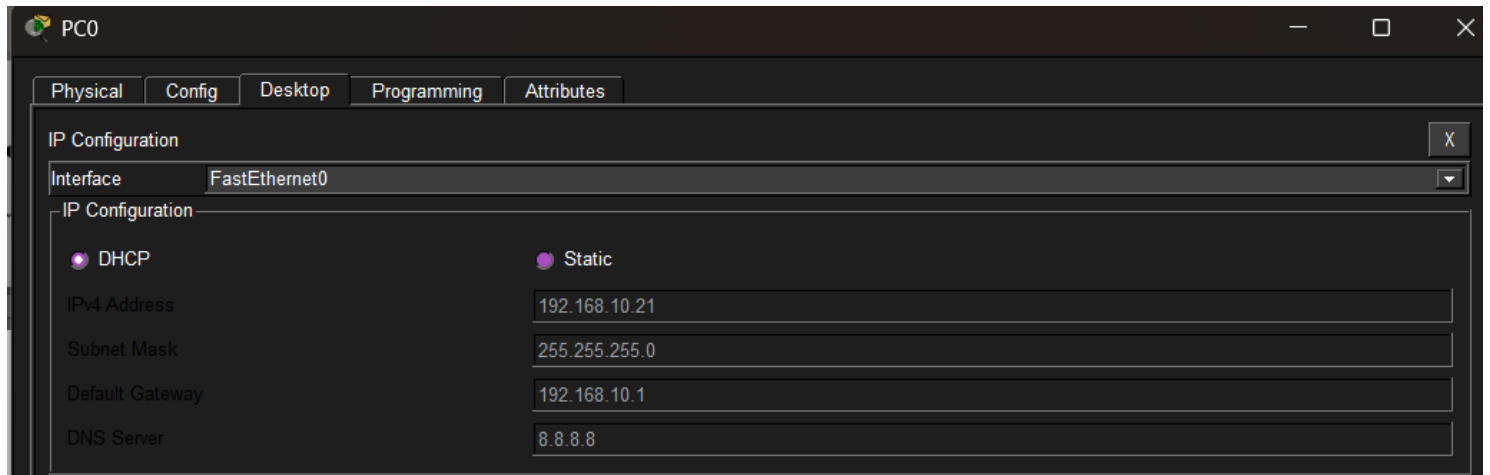
Configure the PCs

For each PC:

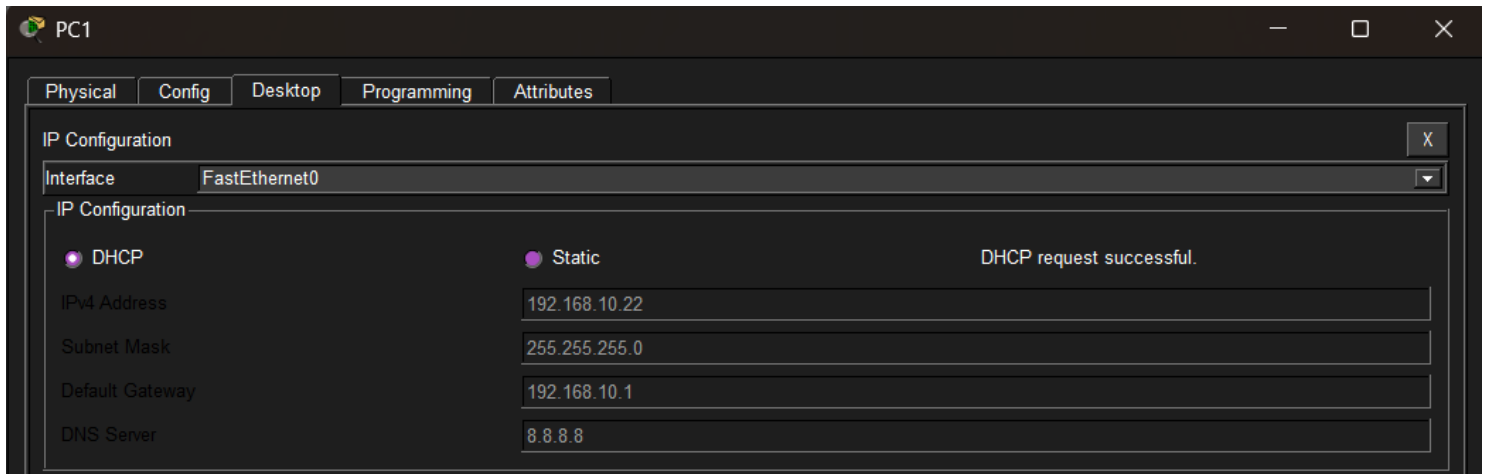
1. **Open** Desktop.
2. **Select** IP Configuration.
3. **Click** DHCP.

The PCs should automatically receive IP addresses.

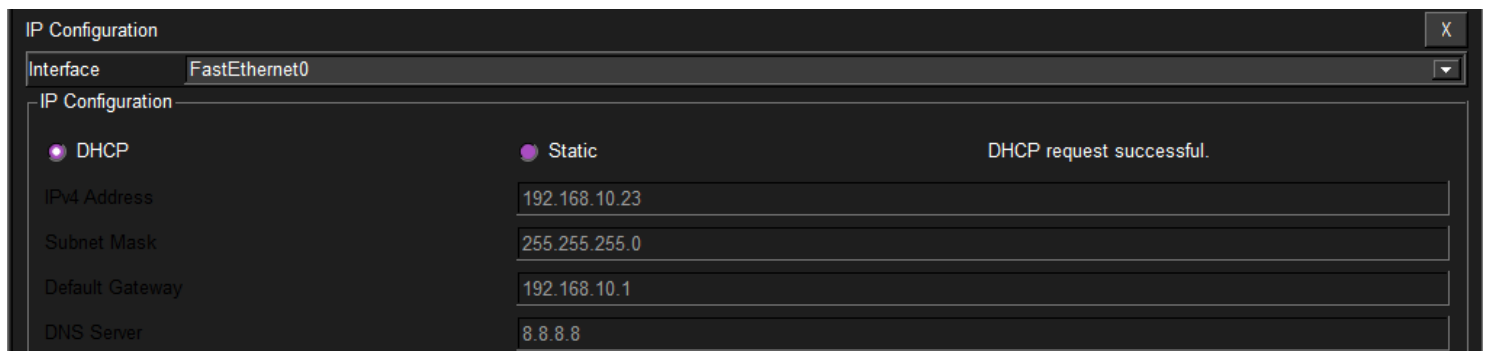
PC0:



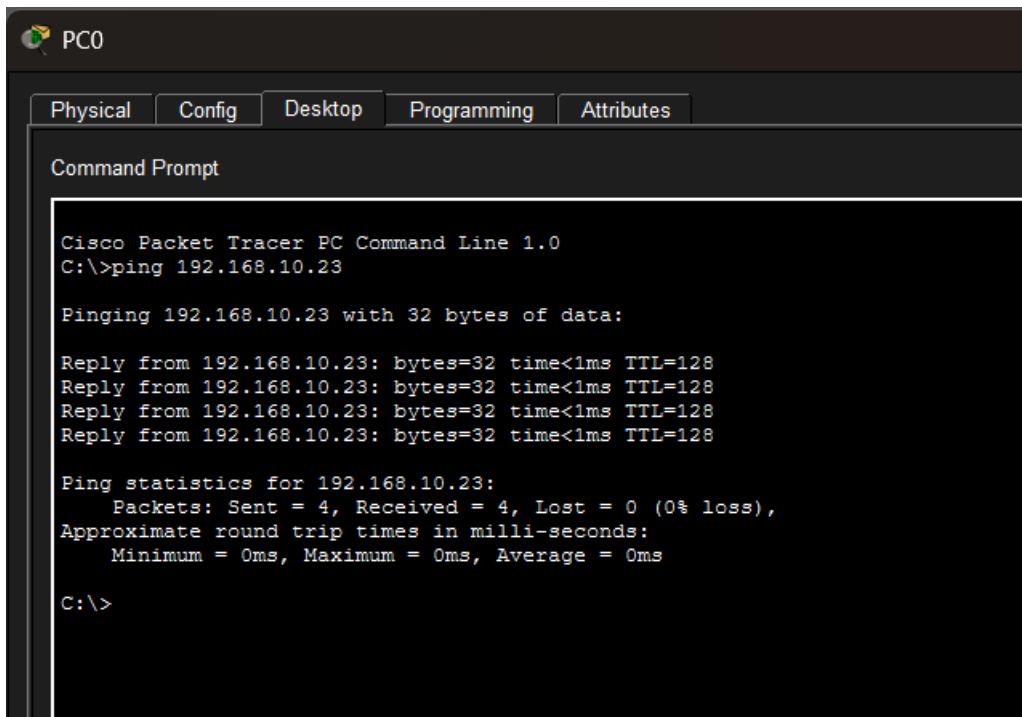
PC 1:



PC 2:



Test connectivity:



The screenshot shows the PC0 configuration window in Cisco Packet Tracer. The 'Config' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the output of a ping command to 192.168.10.23, showing four successful replies with 0% loss.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.23

Pinging 192.168.10.23 with 32 bytes of data:

Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.23:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```